

Shih-Ming Huang

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Specialization

Antenna Design and Measurements / Phased Arrays / RF System Design / RFIC

Skills

- ♦ **Electromagnetic Simulation Software**

Ansyz HFSS, CST Studio, Ansys Maxwell, Keysight ADS, Sonnet, and Altair Feko,

- ♦ **Microwave Devices Measurement**

NSI2000 Antenna Measurement System, Vector Network Analyzer, and Spectrum Analyzer

- ♦ **RF System Measurement**

Specific Absorption Rate (SAR) and Rate versus Range (RvR) Test

Education

University of California, Los Angeles

Sep. 2022 – Now

Ph. D in Electrical & Computer Engineering

- Advisor: Dr. Yuanxun Ethan Wang
- Research focus: ferromagnetic resonance enhanced electrically small antennas

National Taiwan University

Sep. 2016 – Jun. 2022

B.S. in Electrical Engineering and M.S. in Communication Engineering

- Advisor: Dr. Shih-Yuan Chen
- Research focus: spatially reconfigurable phased array

Work & Research Experience

- ♦ **Internship in Product RF Design Team @ Apple**

Jun. 2024 – Sep. 2024

- Designed and implemented Wi-Fi antenna modules
- Systematic evaluation of radiation efficiency, SAR, and data rate of Wi-Fi antenna modules

- ♦ **Ferromagnetic Resonance Enhanced Electrically Small Antenna**

Sep. 2022 – Now

- Designed ferromagnetic resonance based electrically small antenna with $ka = 0.176$ at 3.5 GHz
- Simulated and optimized the design with Ansys HFSS and Ansys Maxwell

- ♦ **Spatially Reconfigurable Phased Arrays**

Aug. 2020 – Jun. 2022

- Established a pair of phase-synced wireless modules with 7° phase error at 433 MHz
- Integrated an VNA with an electronic slider for real-time S-Parameter measurement

- ♦ **A Dual-Band Wearable Open-Sourced Radar System**

Jul. 2019 – Jul. 2020

- Designed and implemented an open-source FMCW radar using 5.8-GHz and 915-MHz ISM band

Publication

- [1] S.-M. Huang, Y.-T. Tsai, H. Chen, C.-S. Shen, L.-Y. Kung, S.-Y. Chen, "[ArrayEd: A simple beam-visualization system for understanding phased array](#)" in *IEEE Antenna and Propagation Magazine*, vol. 66, Issue: 6, 2024.
- [2] R.-F. Xu, L.-C. Ippet-Letembet, S. Tiwari, Z. Yao, S.-M. Huang, Rob N. Candler, S.-Y. Chen, "[Experimental validation of multiferroic antennas in GHz frequency range](#)," in *Applied Physics Letters*, vol. 123, 2023.
- [3] S.-M. Huang, Joshua Yu, Y.-E. Wang, "[An Electrically Small Antenna with Circularly Polarized Electron Spin Precessions \(E-Spin\)](#)," in *2024 IEEE International Symposium on Antennas and Propagation*.
- [4] S.-M. Huang, Q. Gao, Y.-E. Wang, "[A Ferromagnetic-Resonance Based Electrically Small Antenna with Circular Polarization](#)," in *2023 IEEE International Symposium on Antennas and Propagation*.
- [5] S.-M. Huang, W.-C. Chen, Y.-T. Tsai, E. F. Wu, S.-Y. Chen, "[UMPS: Ultrasound-Microwave-Fused Phase Synchronization for UAV-Based Phased Arrays](#)," in *IEEE Asia-Pacific Microwave Conference*, 2021.

Awards

- ♦ **2020 IEEE AP-S Student Design Contest — 1st Place**

International student design contest held by IEEE Antennas and Propagation Society (AP-S)